

Urban sustainability, seen from orbit

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SESAC Beginner Earth Observation Analysis Series



How many people live in cities?

Take a moment. Commit to a number before I put one on the screen.

Three answers. All published. All correct.

THE 2025 ANSWER

45%

live in cities, under the UN's new
grid-based classification

THE OLD ANSWER

58%

live in urban areas, adding up
national definitions

THE OTHER 2025 ANSWER

81%

live in cities or towns, under the
same new classification

Three different rules were applied.

United Nations Department of Economic and Social Affairs, World Urbanization Prospects 2025., 2025.

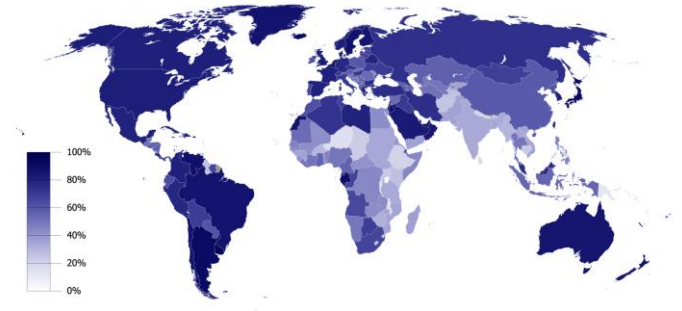
Urbanization and Population Growth

From 55% currently, by 2050, two thirds of the world's population will live in urban areas

Cities consume more than 75% of the world's used resources

Cities contribute to 70% of global CO2 emissions

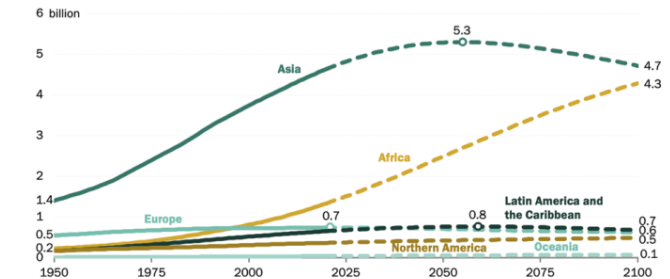
From an urbanization rate of **15 per cent in 1960** to **40 per cent in 2010**, it is projected that **60 per cent of Africans** will live in cities by **2050** (UN Habitat 2014).



2018 Revision of World Urbanization Prospects, UN /Map created by Arthur Gunn <http://gunn.co.nz/map/>

Population growth in Africa is projected to remain strong throughout this century

Population by region, in billions



Note: Data labels show projected peak population for each region: Europe (2021), Asia (2055) and Latin America and the Caribbean (2058). Regions follow United Nations definitions and may differ from other Pew Research Center reports. Source: United Nations Department of Economic and Social Affairs, Population Division, "World Population Prospects 2019."

For seventy years, “urban” meant whatever each country said it meant

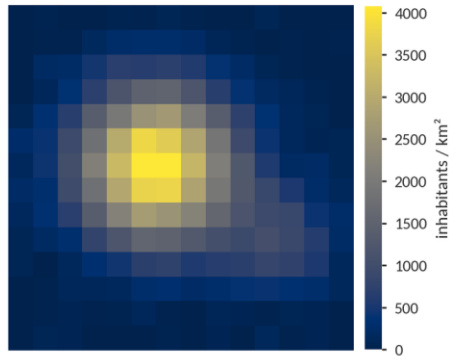
- Every national statistical office sets its own rule such as a population threshold, an administrative status, a density, a share of non-agricultural employment.
- Those rules were never designed to be comparable with each other.
- Adding them up gives 58% urban: an arithmetic total of incompatible definitions, not a measurement of the same thing in every country.
- The mismatch is systematic. Settlements the new classification calls towns are counted as rural in many national systems, particularly across Central and Southern Asia.

A comparative urban statistic needs a rule that does not stop at the border.

The new answer starts from a population grid, not a boundary

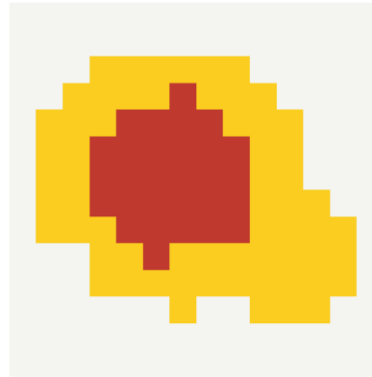
The Degree of Urbanisation: one square-kilometre grid, two thresholds, applied identically in every country.

1. A population grid



1 km cells. Density, not borders.

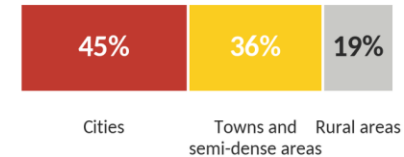
2. Two thresholds



$\geq 1500 / \text{km}^2$ and $\geq 50\,000$ people → city
 $\geq 300 / \text{km}^2$ and $\geq 5\,000$ people → town
everything else → rural

3. The same rule, everywhere

8.2 billion people, 2025

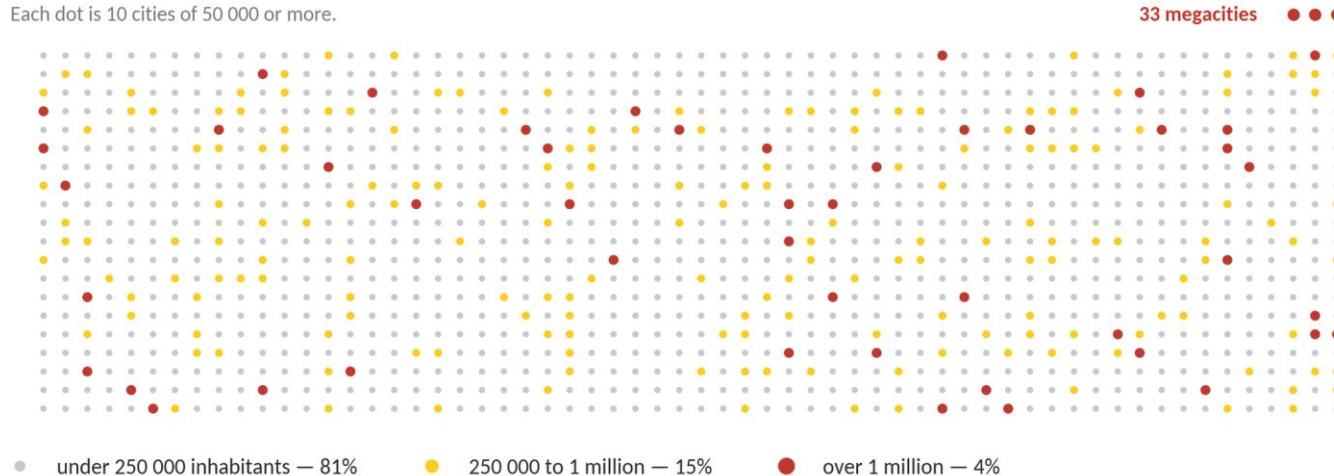


**National definitions give 58% urban.
Same world, different rule.**

Endorsed by the United Nations Statistical Commission at its 51st session. Built on the Global Human Settlement Layer, GHS-POP and GHS-BUILT-S R2023A, derived from Sentinel-2 and the Landsat archive.

Ninety-six per cent of the world's cities have fewer than a million people

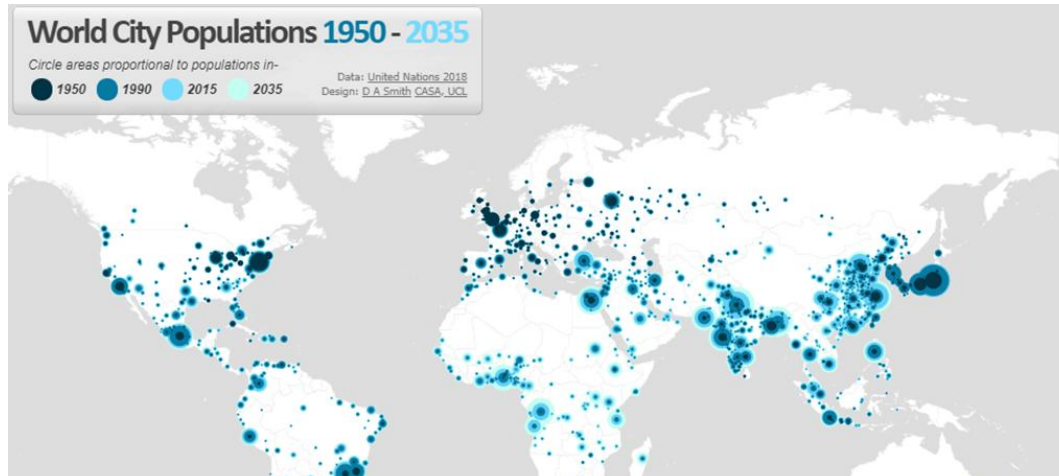
Lowering the size threshold from 300 000 to 50 000 inhabitants took the world's flagship urban database from about 1 900 cities to over 12 000.



Megacities went from 8 in 1975 to 33 today, and are projected to reach 37 by 2050. They are 33 out of more than 12 000.

World Urbanization Prospects 2025.

The growth is not where the attention is



Around 90% of the population increase to 2050 will happen in low- and middle-income cities — and mostly in their deprived urban areas.

World city populations, 1950–2035; circle area is city population.

Ouagadougou

October 2002



Google earth

200 m



Ouagadougou

January 2010

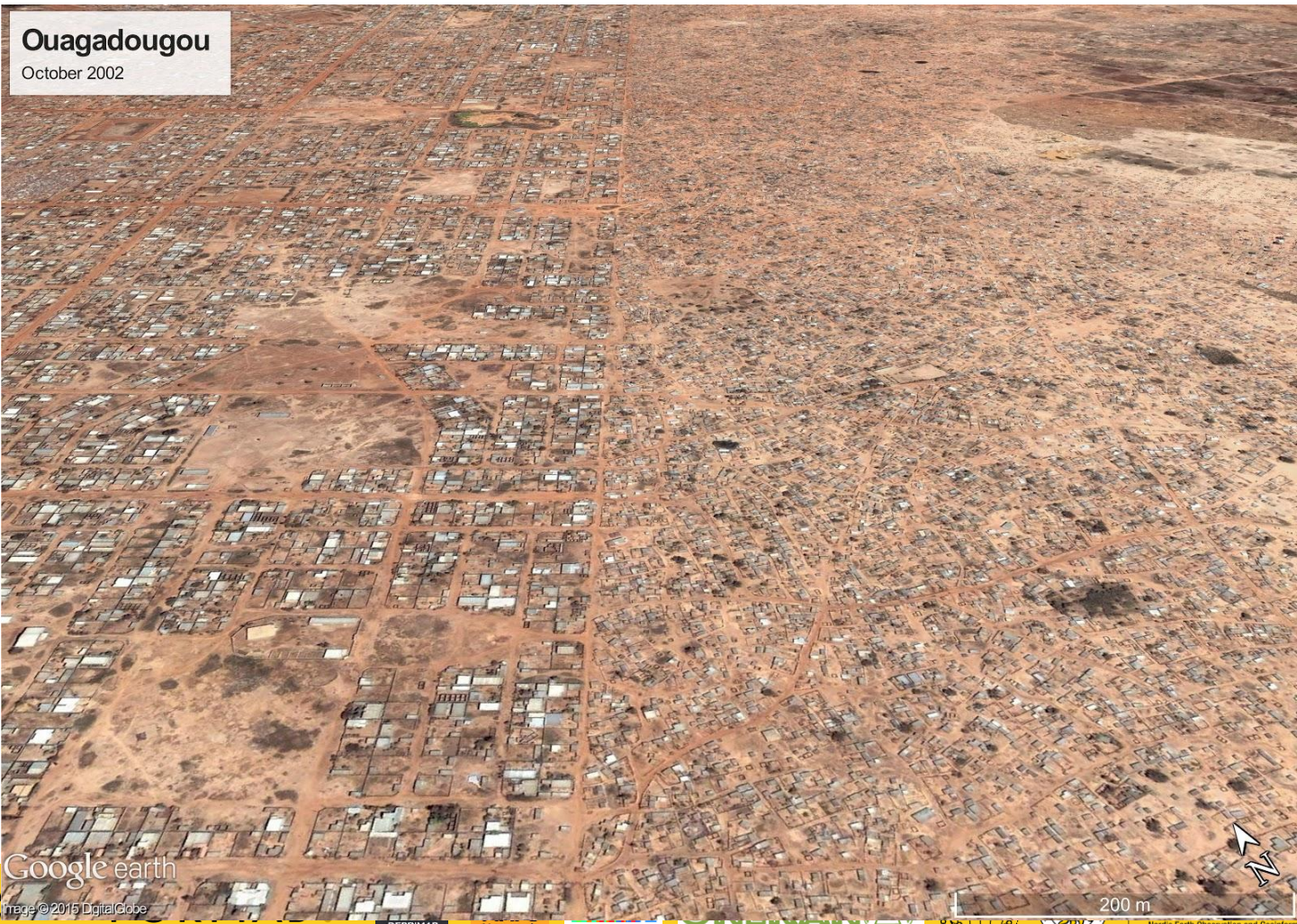


Google earth

200 m

Ouagadougou

October 2002



Google earth

Image © 2015 DigitalGlobe

200 m



Ouagadougou

October 2009



Google earth

200 m



2002

Google Earth

Image © 2020 Maxar Technologies
Image NASA

500 m

2018



Google Earth

Image © 2020 Maxar Technologies,
Image NASA

What changed was not the sensor. It was the unit of analysis.

- The satellites that made this possible have been flying for years. Sentinel-2 and the Landsat archive are not new, and nothing about them was improved for this.
- What was new was agreeing on a rule, one grid, two thresholds that could be run identically in 194 countries.
- That is the first and least glamorous contribution of Earth Observation to urban sustainability: not a better picture, but a common denominator.

Now the harder question. We can compare cities with each other. Can we see anything inside one?

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Enter the code

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Module 01: What counts as urban?



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A spatial problem, measured by a household survey, reported by country

SDG 11.1.1 — the proportion of the urban population living in slums, informal settlements or inadequate housing.

SDG 11.1.1, 2026

1.13 bn

people in slums and informal
settlements

CHANGE SINCE 2015

+130 M

more people, not fewer, since
the target was adopted

INADEQUATE HOUSING

1.6 bn+

people worldwide, on
UN-Habitat's wider measure

—The indicator is a household attribute such as water, sanitation, durability, overcrowding, tenure aggregated to a national percentage.

— A national percentage cannot tell a municipality which streets to work on.

United Nations Statistics Division, Extended Report 2026.

In Nairobi, most of the people live on a small fraction of the built-up area

SHARE OF RESIDENTS

~60%

of Nairobi's inhabitants live in
deprived urban areas

SHARE OF BUILT-UP AREA

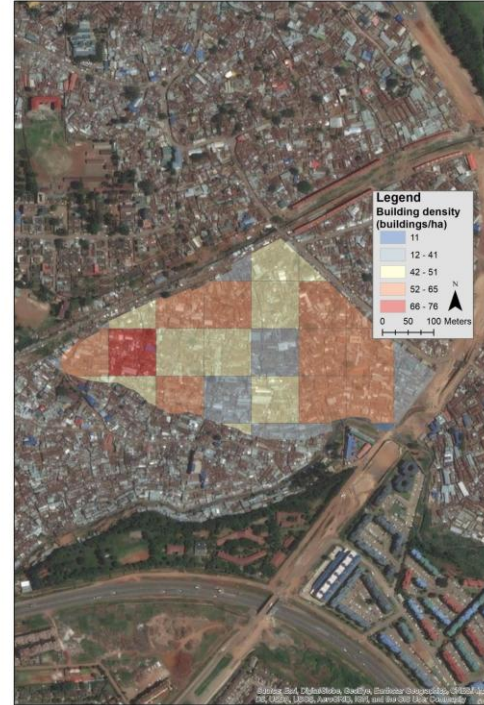
~4%

of the city's built-up surface is
what they live on

GLOBAL POPULATION MODELS

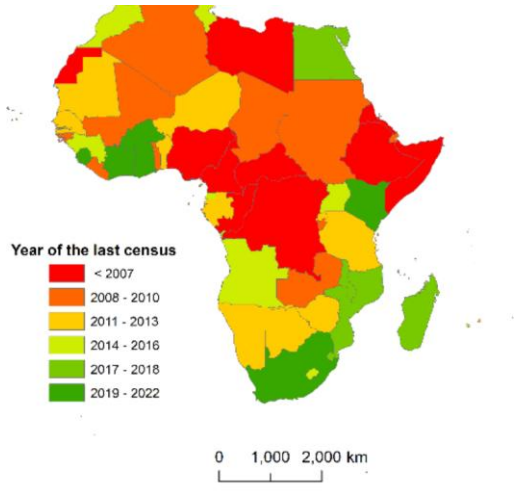
<10%

of that population is sometimes
all a global product captures



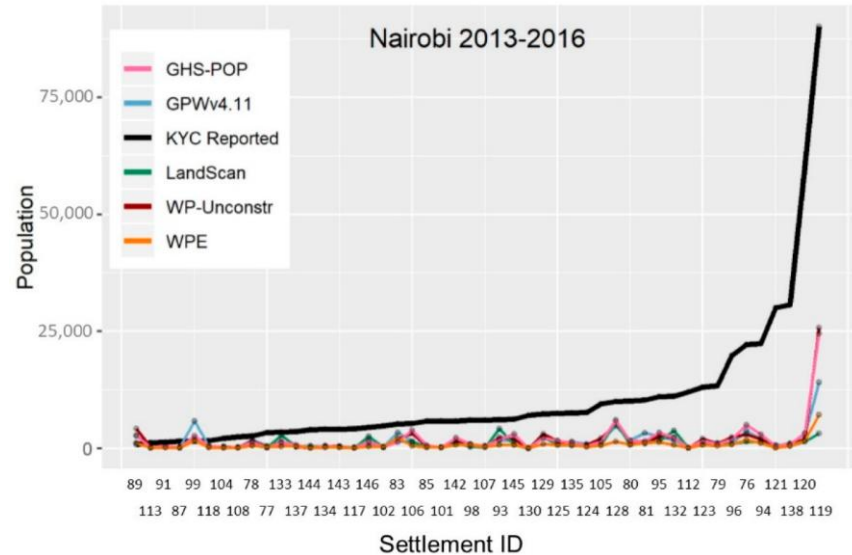
A model that is 90% right about the city can still be almost entirely wrong about the people in it.

The data underneath is oldest and thinnest exactly where the problem is worst



Years since the last census occurred in African countries (reference year 2022)

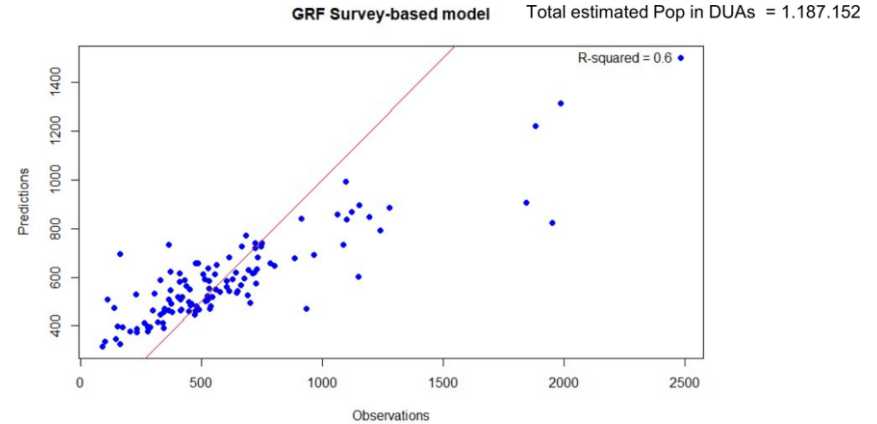
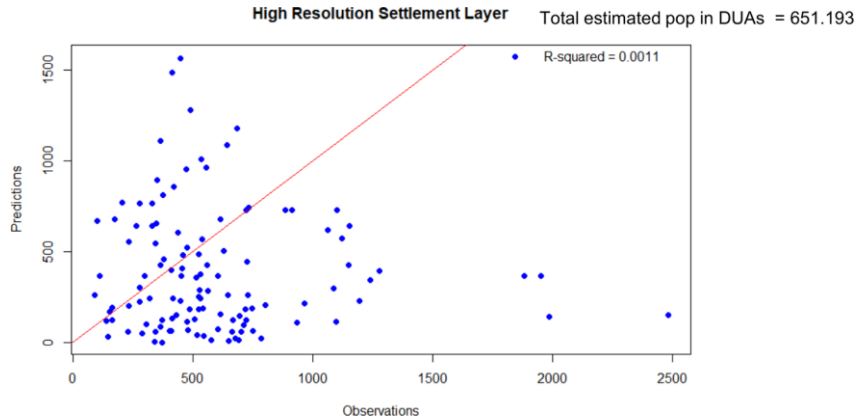
Left: years since the last census, African countries, reference year 2022



Right: gridded population products against reported settlement counts, Nairobi 2013–2016. Thomson, D.R. et al. (2021), Urban Science 5(2):48.

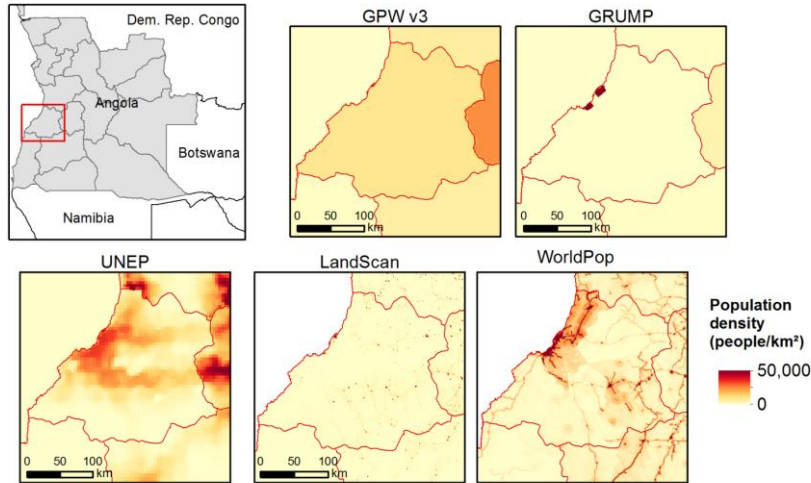
Same city, same ground truth, two models, half a million people apart

Both panels validate against the same survey measurements in Nairobi's deprived urban areas.

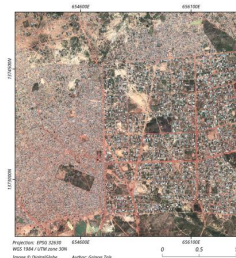


The people missing from the data are the people the policy is for.

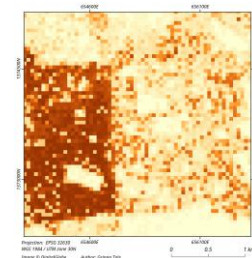
Where a “population map” actually comes from



Five global gridded population products, same region



A satellite image ...



... redistributed into a density grid

— None of these products counts anybody. Each takes a census total and redistributes it using satellite-derived weights, built-up surface, roads, land cover, night lights.

— So the census is upstream of every one of them. Where the census is old, coarse or incomplete, the grid inherits all three problems and adds a modelling error of its own.

Module 02:

The data gap

inside cities



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Morphology is the spatial imprint of disadvantage.

So: what is the imprint actually made of, and how much of it can a satellite resolve?

Which parts of the definition can a satellite actually reach?

UN-Habitat's criteria for slums, informal settlements and inadequate housing, set against what Earth Observation can supply.

Access to water and sanitation

Partly — unplanned urbanisation as a proxy

Sufficient living area, overcrowding

Yes — built-up density and building size

Structural quality and durability

Yes — roofing material and location

Security of tenure

Partly — unplanned urbanisation as a proxy

Accessibility

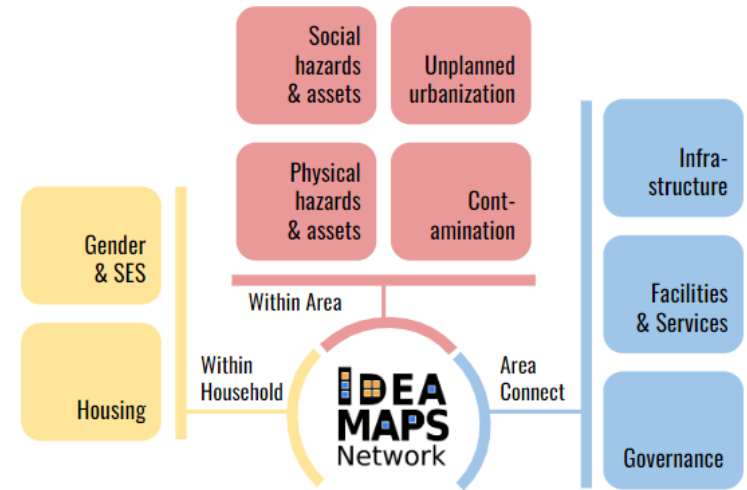
Yes — road network and geospatial data

Affordability

No — requires local data

Deprivation is nine domains at three scales

- The household scale, income, tenure, who lives here is survey territory.
- The area scale, how the place is built, what is in it, what it is exposed to is where Earth Observation works.
- The connection scale, what the area can reach, is geospatial, part EO and part network data.



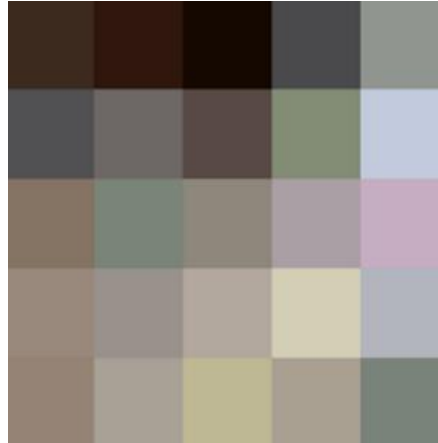
Domains or Deprivation paper:

<https://doi.org/10.1016/j.compenvurbsys.2022.101770>

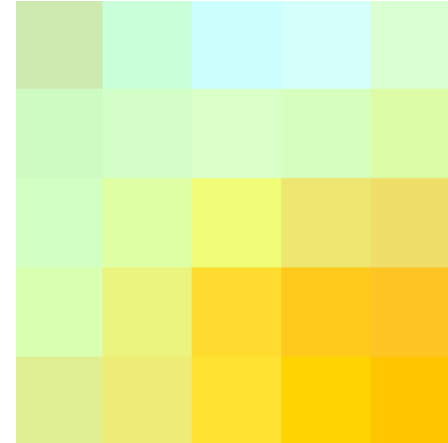
The same fifty metres of Nairobi, through three sensors



Very high resolution optical
~0.5 m — individual roofs



Sentinel-2
10 m — texture, not buildings



Sentinel-1 radar
10 m — roughness, all weather

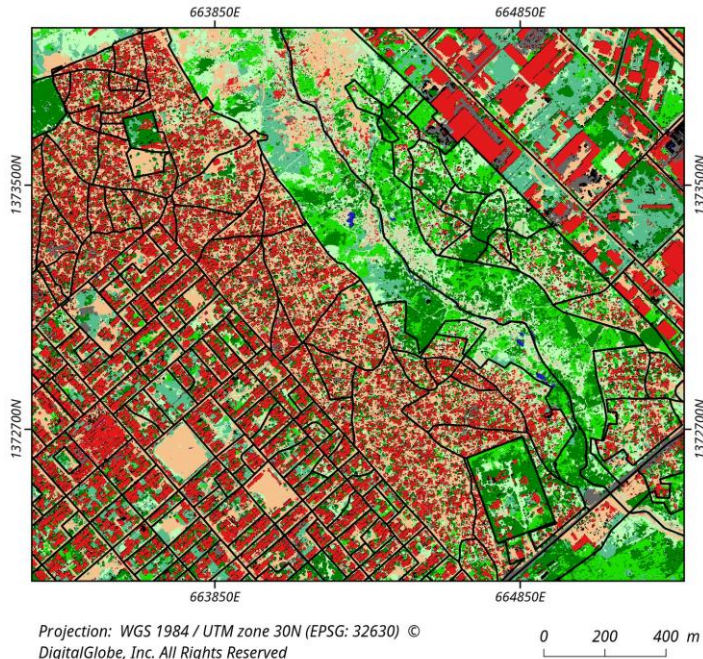
Free, global, every few days — but at 10 m a deprived block is a texture, not a set of houses. Almost everything that follows is built from that texture.

Some of the imprint needs no model at all

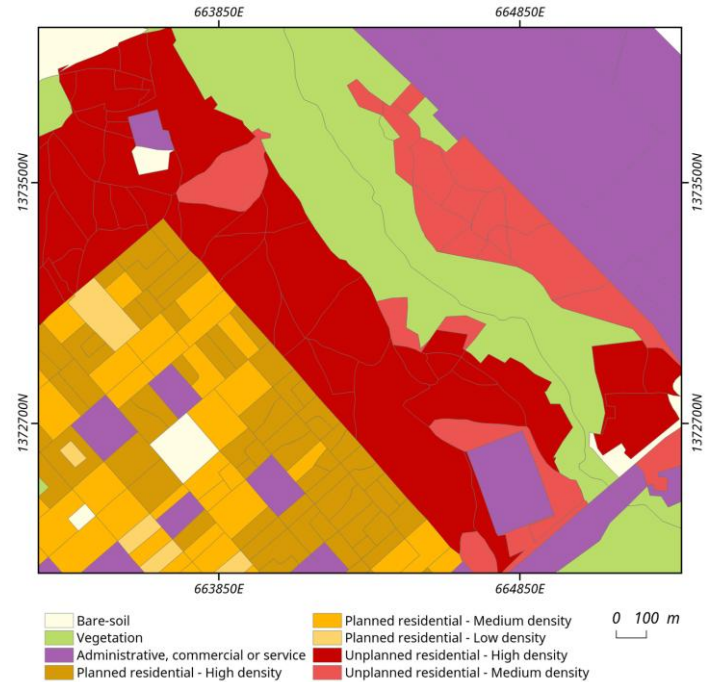


Inequality has an edge, and the edge is a physical object, a road, a wall, a drainage line. That is why it can be measured from above at all.

Domain one — unplanned urbanisation: built form alone separates the two cities

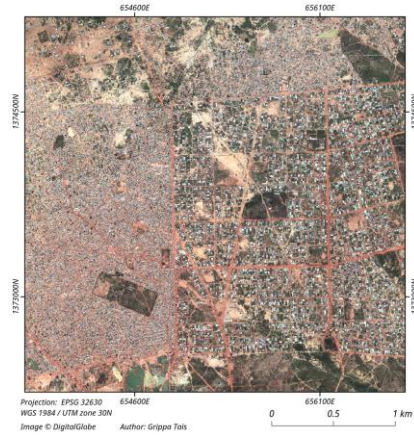


Land cover in street blocks

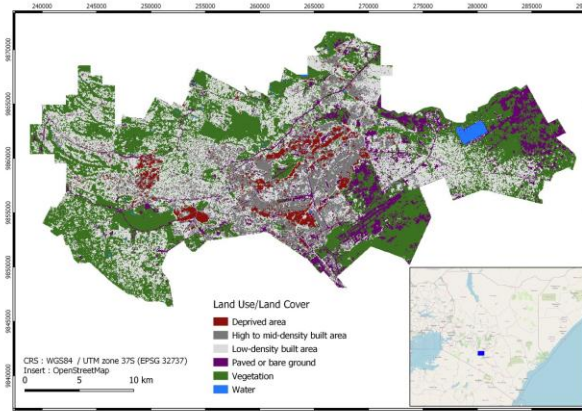


Land use, classified per block

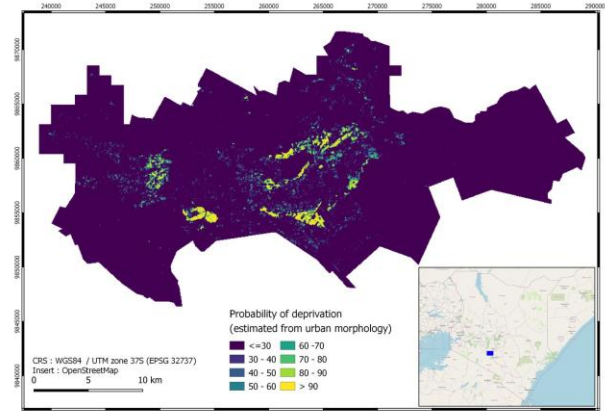
An image becomes a probability of deprivation



1 · Very high resolution image



2 · Land use and land cover, classified



3 · Probability of deprivation, from morphology

But deprived areas are not one kind of place



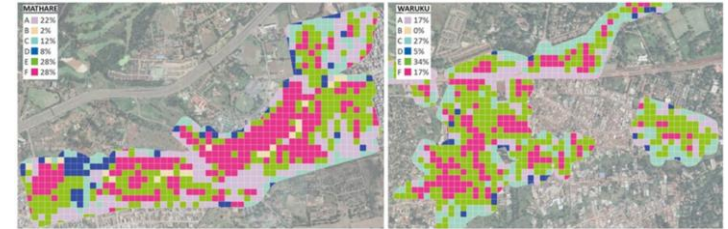
Typical



Both photographs are deprived areas in the same city. A model trained on the first will find the first; whether it finds the second is the whole question of transferability.

Typology

CLUSTERS		LC FEATURES								DESCRIPTION
		Waste Piles	Building	Low Vegetation	Tall Vegetation	Vehicles	Shadow	Ground Surface	Water	
A	→									High Ground Surface coverage; Medium Building Density; Some presence of Waste Piles and Vehicles.
B	→									High presence of Water bodies; High presence of Waste Piles; Some Tall and Low Vegetation; Medium-Low Building Density.
C	→									High presence of Tall Vegetation; High presence of Shadows; Some presence of Low vegetation and Low Building density.
D	→									High Low Vegetation coverage; Some Tall Vegetation; Some presence of Waste Piles.
E	→									High-Medium Building Density; High presence of Shadows; Little Low and Tall Vegetation; Some Waste Piles and Vehicles.
F	→									High Building Density; Medium presence of Shadows; Some presence of Vehicles; Lack of Vegetation.



Six clusters, each with a written description high ground surface, medium building density, some presence of waste piles and vehicles instead of one binary called “slum”.

Nairobi, WorldView-3 superspectral imagery. Georganos, S., Abascal, A., Kuffer, M., Wang, J., Owusu, M., Wolff, E. & Vanhuyse, S. (2021). Is It All the Same? Mapping and Characterizing Deprived Urban Areas Using WorldView-3 Superspectral Imagery. Remote Sensing 13:4986.

Module U3: Reading deprivation from above



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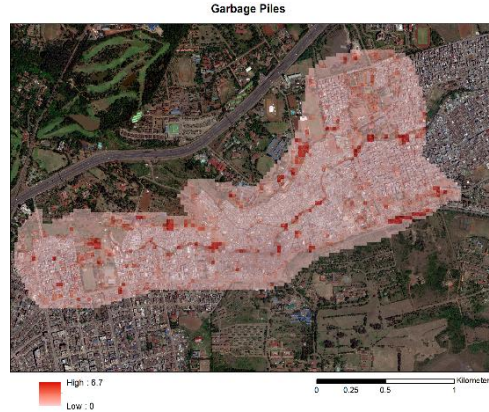
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Waiting for participants

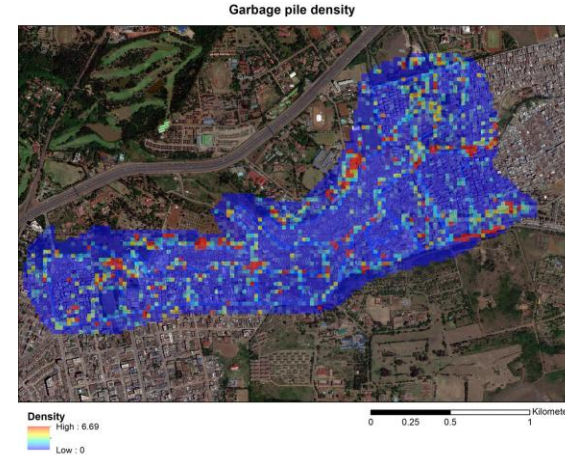
Contamination: waste is large enough and bright enough to count



Mathare river, Nairobi



Detected garbage piles



Pile density across the settlement

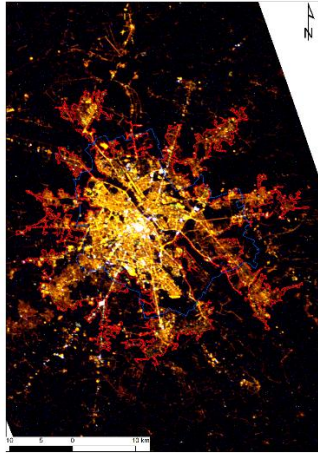
Solid-waste accumulation is an SDG 11.6 indicator that almost no city measures. From very high resolution imagery it becomes a map, not an anecdote.

What the detection actually looks like

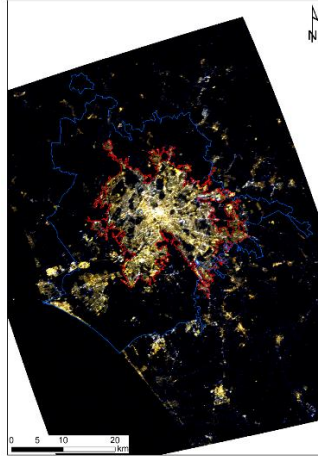


Buildings, vegetation, bare ground, asphalt, water, vehicles — and garbage piles as a class of their own, mapped object by object.

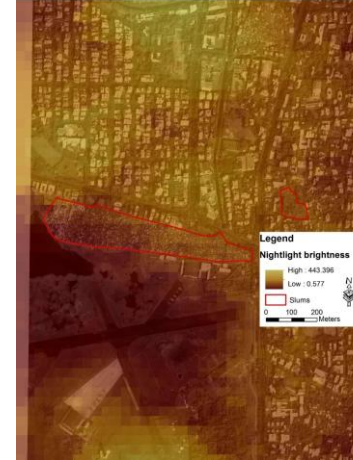
Infrastructure: at night the city shows you its grid connection



Rome — astronaut photography



Warsaw — built-up outline overlaid

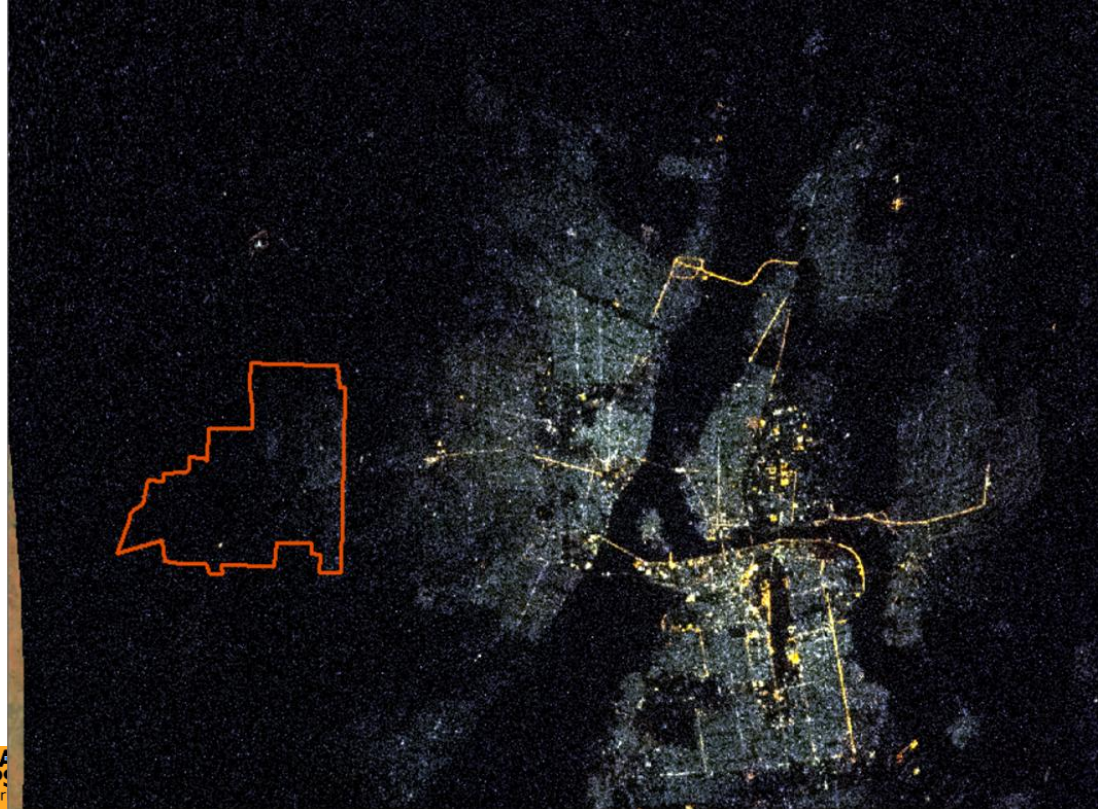


Brightness against settlement boundary

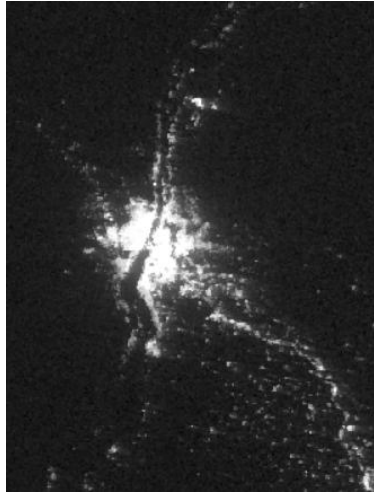
SDG 7.1.1 asks what proportion of people have electricity. Night lights answer a sharper question: which streets, and on which nights.

Missing Street Lights

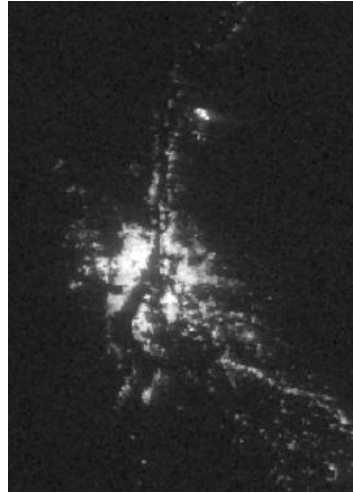
Khartoum 1985 - 2015



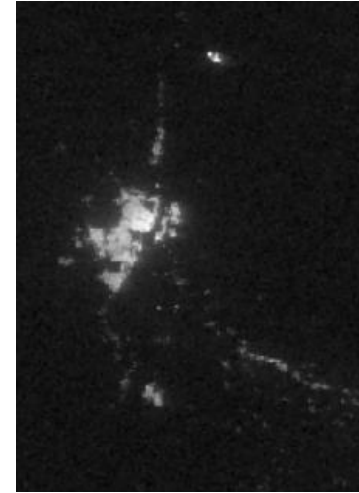
And when a city goes dark, the satellite is often the only witness left



Khartoum, March 2023



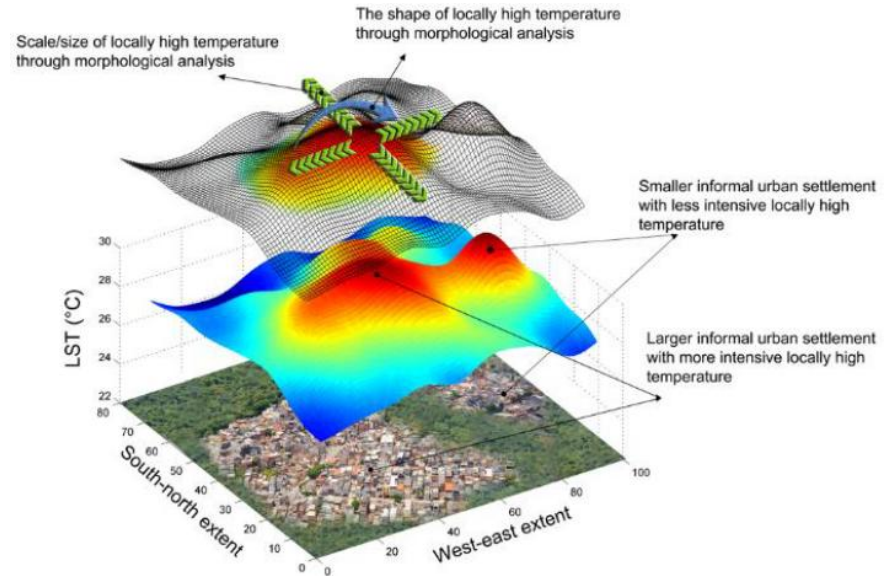
Khartoum, May 2023



Khartoum, August 2023

Environment: the same built form also traps heat

- Dense, low, unshaded built form with little vegetation produces the highest local surface temperatures and it is the same built form the deprivation models key on.
- So heat exposure and morphological deprivation are not two separate maps. They are largely the same map, read with a different band.
- In European cities, land surface temperature during extreme events can run 10 to 15 °C above the surrounding countryside.
- This is the domain that transfers most directly to Sweden



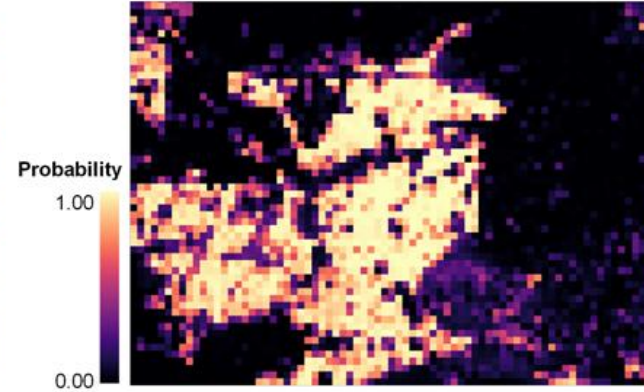
Deprivation and LST

WARUKU
(KAWANGWARE)

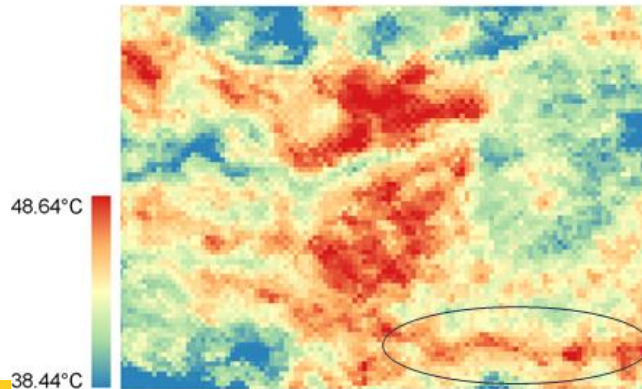
SENTINEL-2 RGB



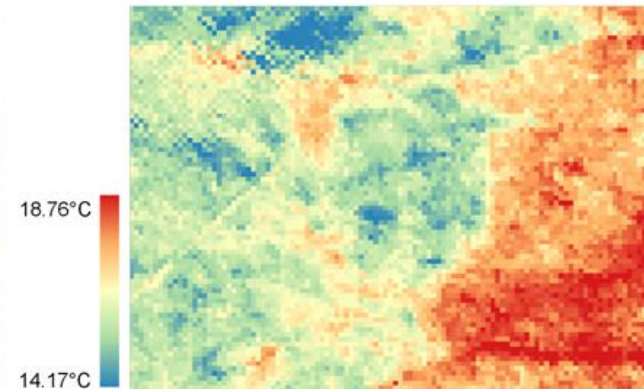
MORPHOLOGICAL DEPRIVATION



ECOSTRESS LST 28/02/2021 DAYTIME



ECOSTRESS LST 04/02/2021 NIGHTTIME



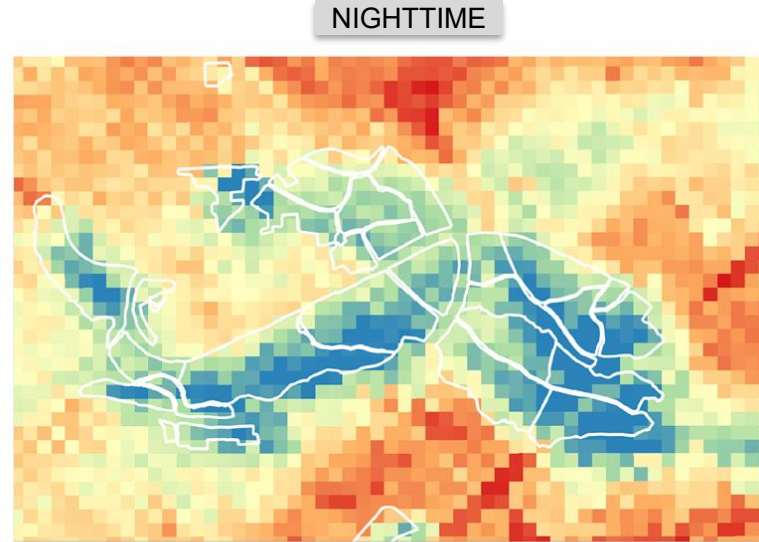
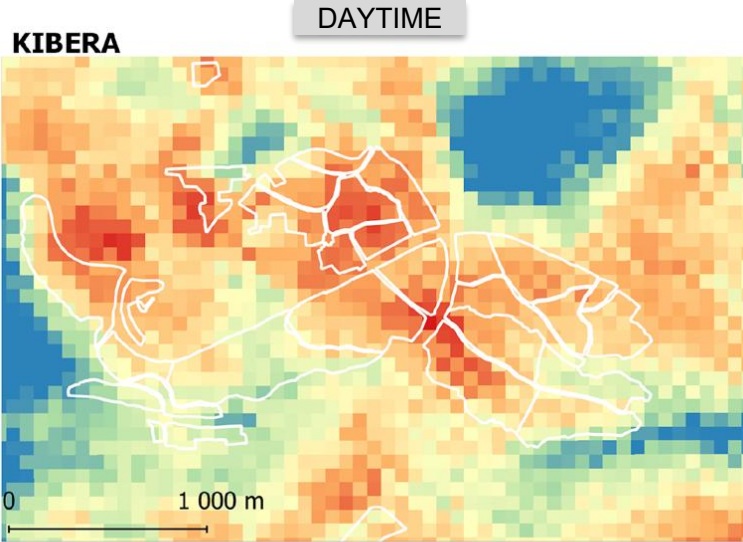
0 1000 2000 3000 4000 m

Deprivation and LST



Metal sheet roofs

- Relatively high LST during daytime
- Relatively low LST during nighttime



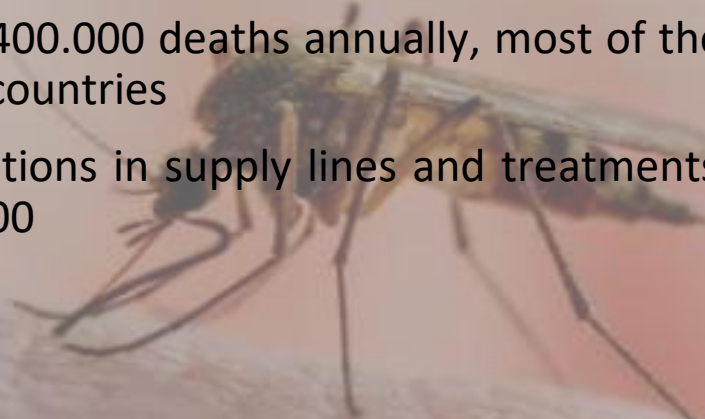
LST

Low

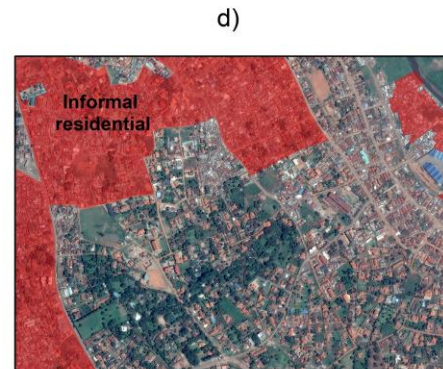
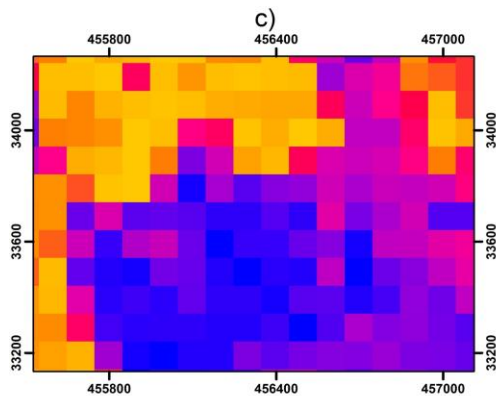
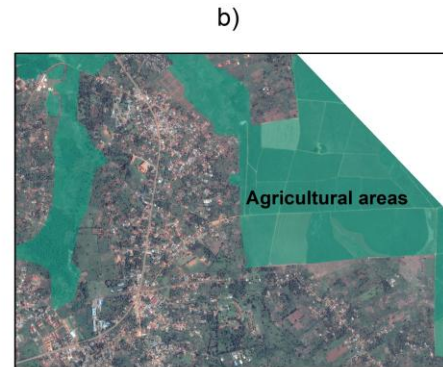
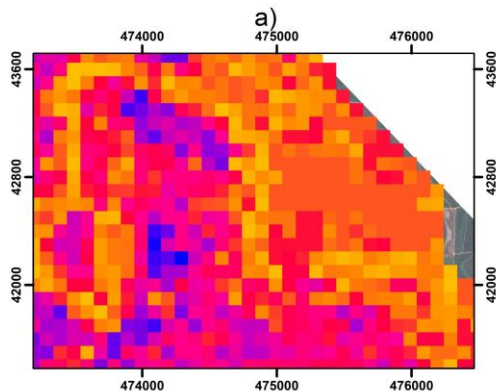
High

Urban deprivation and epidemiology

- One of the most prominent use of RS information is related to spatial epidemiological models
- Malaria is an infectious disease caused by the *Plasmodium* parasite, transmitted to humans through the *Anopheles* mosquito vectors
- It causes more than 400.000 deaths annually, most of them children aged under 5-years old and in SSA countries
- COVID-related disruptions in supply lines and treatments may increase the annual fatalities up to 715.000



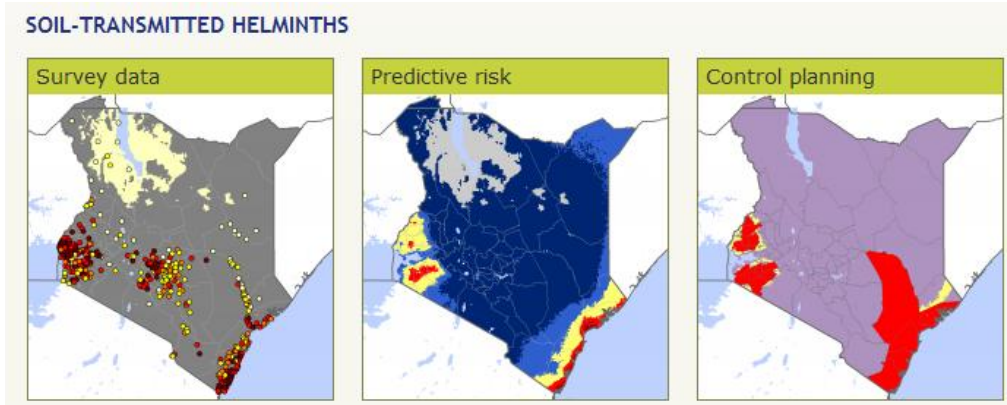
Urban deprivation and epidemiology



PPR2_10
High : 15.21
Low : 2.59

Projection System: WGS 1984, UTM 36N

The point of a risk map is the decision that follows it



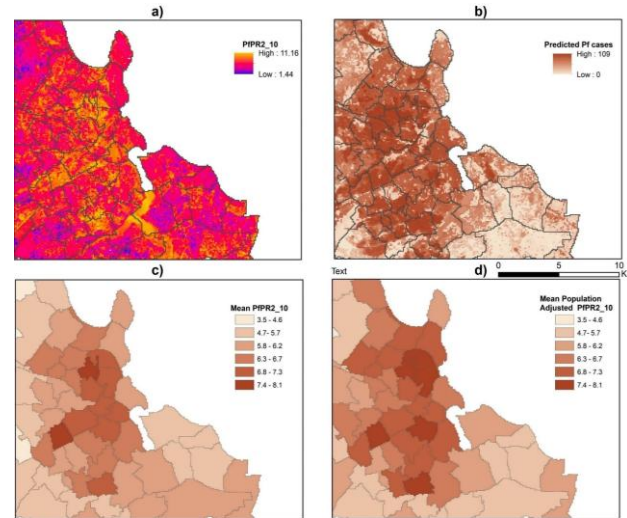
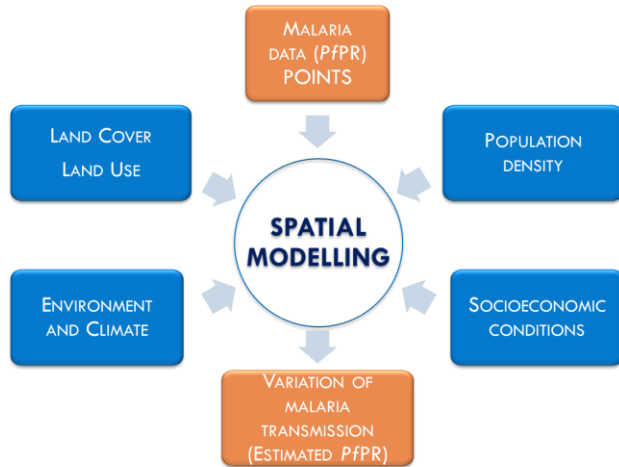
Survey data → predictive risk surface → control planning. Soil-transmitted helminths, Kenya.



Modelled mosquito larval habitat suitability, Dakar

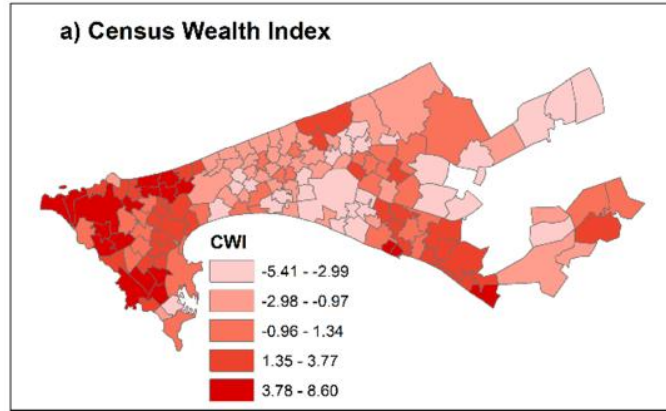
The map is not the output. The output is a list of wards to treat first, and an argument for why those wards.

Domain six — health: built form predicts where transmission concentrates

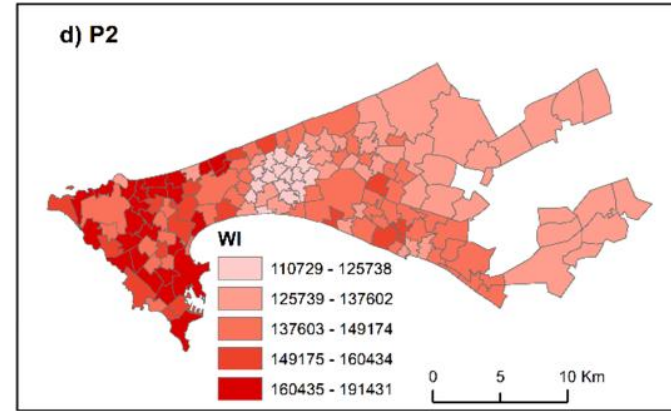


Prevalence rises with dense unplanned settlement and falls with planned neighbourhoods · land-use features are the most explanatory variables · roughly 40% of the variance captured.

Wealth



Census wealth index, Dakar

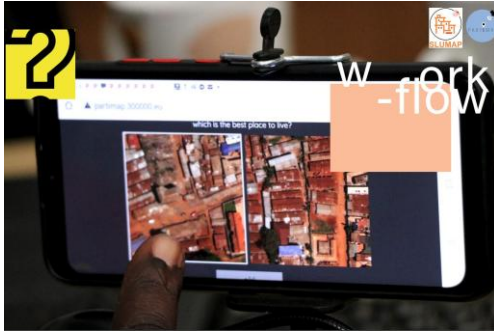


Modelled from satellite-derived indicators

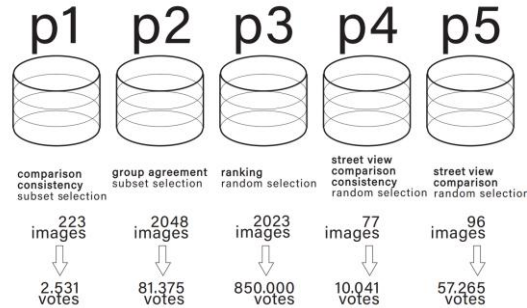
Moderate correlation, credible spatial pattern, and interpretable relationships swimming pools and tall vegetation up, bare ground and dense low building down. It is a prediction of a survey, not a measurement of a household.

Georganos, S. et al. (2019), *Remote Sensing* 11(21):2543. Figures carried from S. Georganos's own decks.

And lately we have started asking people what they see



"Which is the better place to live?"



850 000 pairwise comparisons, five experiments



The citizen-perceived deprivation surface

Whose perception, though — the people who live there, or people looking at pictures of where they live? Those give different answers, and which one you want depends on what the map is for.

The pattern is the same in every domain

Contamination	sees the waste pile	not the collection contract
Infrastructure	sees the dark street	not the unpaid bill
Physical hazard	sees the roof inside the flood band	not the tenure that stops you moving
Environment	sees the hot roof	not who can afford to cool under it
Health	sees the built form that predicts risk	not the bed net, the clinic, the fare
Socio-economic	sees the assets you can see from above	not the income, the debt, the lease

Which is why the interesting question was never “can we see it?”, it was “can we see it the same way everywhere?”

Module 04: The Six Domains



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Waiting for participants

No single way of mapping deprivation is adequate on its own

Field mapping

✓ Most accurate and context-relevant

✗ Does not scale across all cities

Aggregating census and survey

✓ Reflects social characteristics

✗ Slum households are not slum areas

Digitising imagery by hand

✓ Captures local expert agreement

✗ Does not scale across cities

Computer modelling

✓ Scalable across cities, repeatable

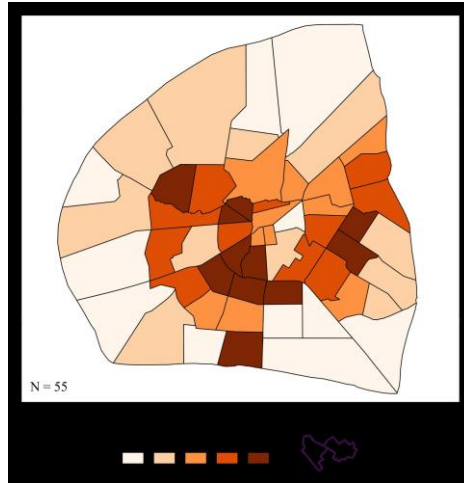
✗ Often divorced from local context

After the IDEAMAPS Network's framing of complementary mapping approaches; carried from M. Kuffer's IDEAMAPS/IDEAtlas deck.

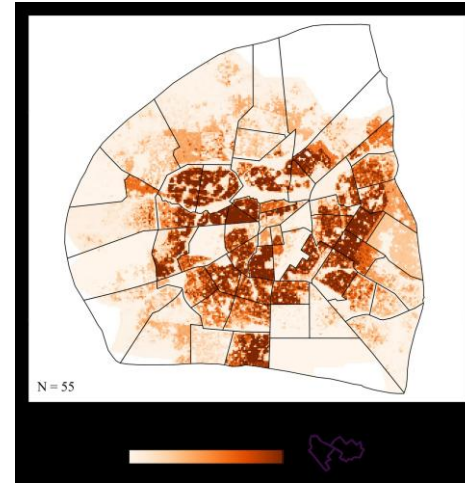
- Open data only. Nothing a municipality would have to buy, and nothing that exists for some cities and not others.
- One spatial unit, constructed identically everywhere.
- One model, trained once. No per-city calibration, no local parameter fitting.
- Then apply it to every city the unit can be built for — and report honestly where it could not be built.

Veeravalli, Blei, Friesen, Tareke, Kuffer, Persello, Maretto, Abascal, Georganos & Thomson (2026). Hidden living-condition deficits in small and medium-sized cities. Nature Cities. doi:10.1038/s44284-026-00498-w

The unit you choose decides what you are able to see



55 administrative units, one value each



The same city on a 100 m grid

Same city, same day, same underlying reality. The left-hand map cannot show you a neighbourhood, because it has no neighbourhoods in it.

A grid cell is an artefact of the analyst. A street block is an artefact of the city.

- City segments are contiguous urban polygons bounded by navigable roads and major water bodies — so they approximate functional neighbourhood units.
- Regular grids cut across the morphology they are supposed to measure. Administrative units are not comparable across countries.
- Each segment holds at least 400 residents, which keeps it demographically meaningful.



Built from four open inputs: GHS-UCDB urban extents · OpenStreetMap roads and GeoFabrik water · GHS-POP for the 400-resident floor · Overture Maps building footprints.

The whole method on one diagram

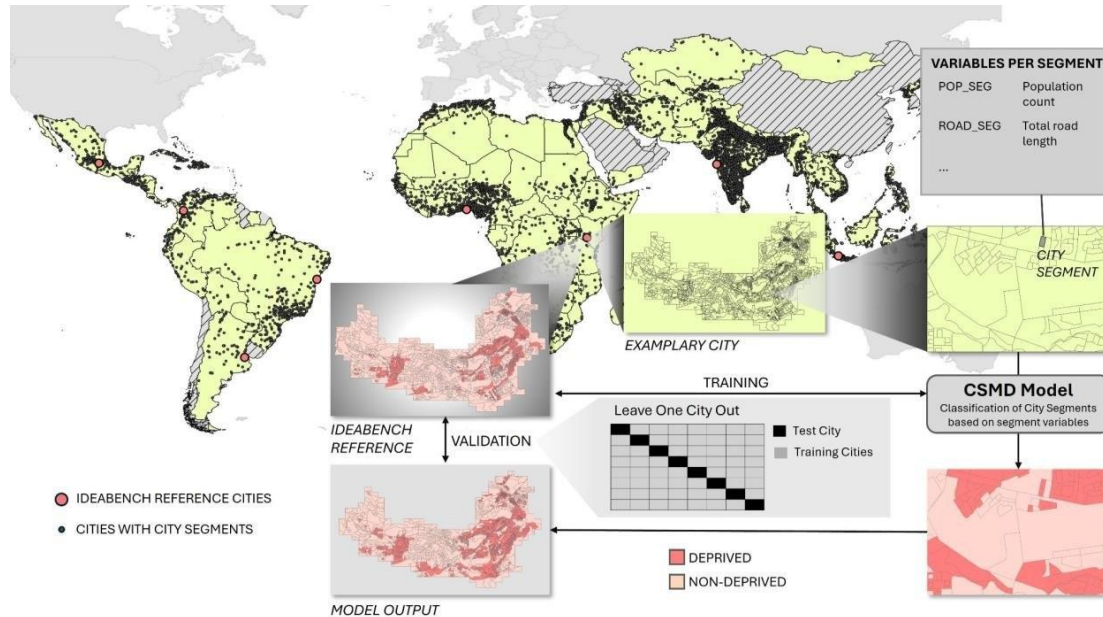


Figure 1, Veeravalli et al. (2026), Nature Cities. Segments and their built-environment indicators are labelled in eight benchmark cities, learned by a random forest, validated leave-one-city-out, then applied to 5 132 cities.

The labels were validated in the field, not drawn in an office

- Reference labels started from datasets held by municipal authorities, community organisations and national statistical agencies.
- Through the IDEAtlas portal, predictions were iteratively checked and corrected with local stakeholders and living labs.
- Eight benchmark cities across three continents: Nairobi, Lagos, Mexico City, Buenos Aires, Salvador, Medellín, Jakarta, Mumbai.



All eight are large cities — roughly 3 to 25 million people. Which is precisely the class of city the headline result is not about.

395 million people, across 5 132 cities in 103 countries

IN DEPRIVED SEGMENTS

395 M

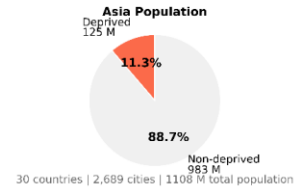
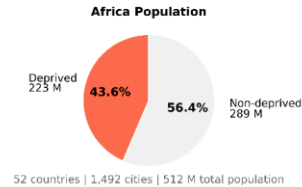
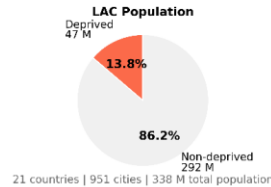
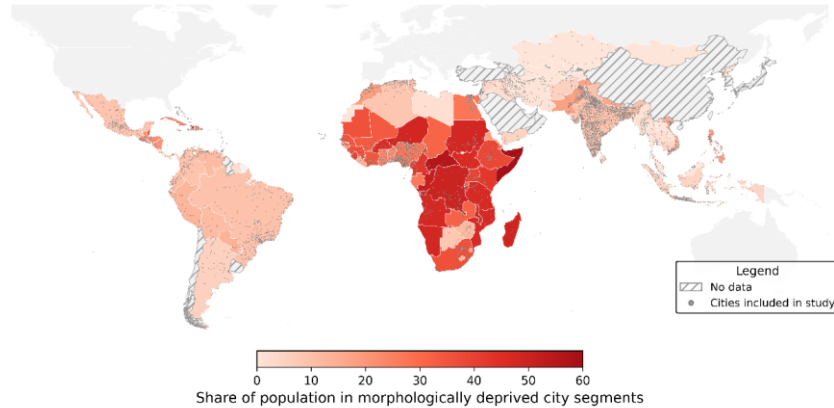
OF COVERED POPULATION

20.2%

URBAN RESIDENTS COVERED

1.96 bn

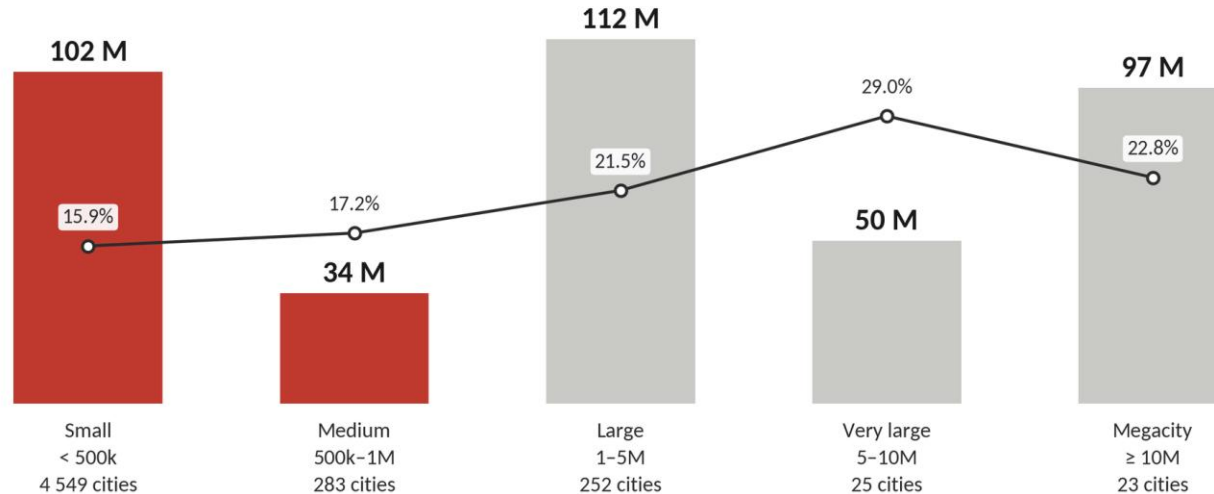
Share of population in morphologically deprived city segments (n=103)



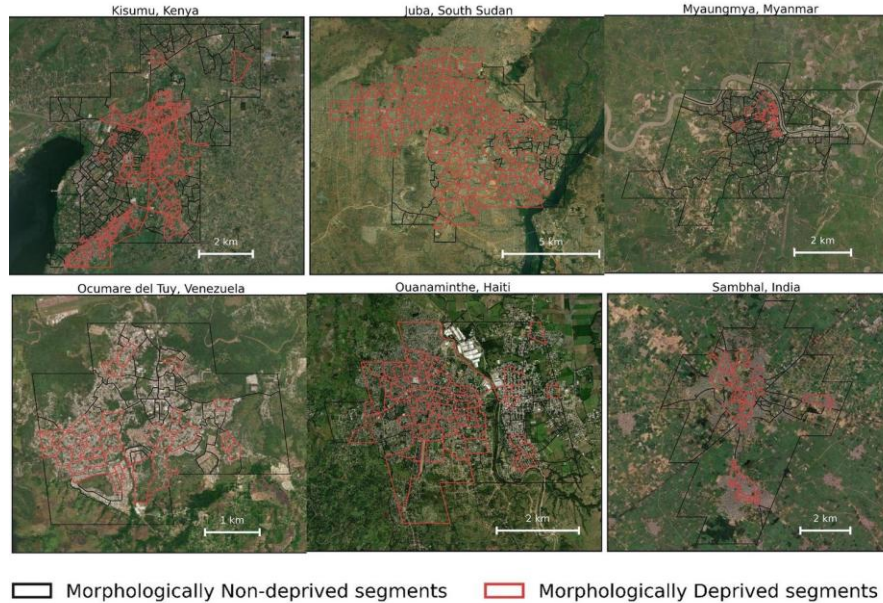
Prevalence rises with city size. The burden does not follow it.

Bars — millions of people in morphologically deprived segments

Line — share of that city class's own population



Six small cities the model had never seen



Extended Data Figure 1, Veeravalli et al. (2026), Nature Cities. Red outlines are morphologically deprived segments; black are non-deprived. None of these six cities is in the eight-city training set.

Publishing a deprivation map has consequences for the people mapped

- A neighbourhood-scale map of where the poorest people live is useful to a planner and useful to a developer, and the same file serves both.
- In several of the cities in this dataset, the documented history of settlement mapping includes eviction.
- The Khartoum sequence you saw earlier required no consent from anyone in it. That is the property that made it possible and the property that should worry us.

Who decides what resolution is safe to release, and to whom? It is a question for this room more than for the modellers.

Sweden has the opposite problem, and it needs the same join

Nairobi, Juba, Sambhal

the problem the model was built for

HAS

A satellite record of every building, road and roof — updated every few days

LACKS

A trustworthy count of who lives there, and how they live

same join



Malmö, Lund, Karlstad

the problem in front of this room

HAS

6 160 DeSO · 700–3 000 residents
3 363 RegSO · register data on income, tenure, health, mobility

LACKS

A current, comparable measure of the physical environment those people actually live in

Three of SDG 11's own indicators are already measured from orbit

11.3.1

Land consumption vs
population growth

2.28%

against 0.87% population
growth, 2020–2025

11.7.1

Urban land that is
open public space

17.0%

UN-Habitat recommends
30–45%

11.6.2

Fine particulate
matter, PM_{2.5}

26 µg/m³

WHO guideline value
is 5 µg/m³

11.1.1

People in slums and
informal settlements

1.13 bn

the one EO can only
partly see

The European layer already exists: Copernicus Urban Atlas 2021, released 2 June 2026, 764 functional urban areas with land use, building height and a street tree layer, now updated every three years instead of every six.

Module 05: Consequences and Sweden



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Three claims to take away

1 **Earth Observation's contribution is a common unit, not a better picture.**

One grid and two thresholds redrew the urban map, and revealed that 96% of the world's cities are small ones.

2 **In every domain, it measures the environment and not the entitlement.**

Waste, light, flood, heat, disease risk, assets, the physical consequence is visible and the social arrangement that produced it is not. Be precise about which half you are supplying.

3 **Deprivation runs the full urban hierarchy, and the output is a scaffold.**

395 million people across 5 132 cities, a third of them in small and medium cities. A screening layer that locates where to look, not a verdict on anyone's street.

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